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Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

ITEP 414 – System Administration and Maintenance

Week 1 Laboratory Activity

Orientation & Building Your Professional Environment

Program: Bachelor of Science in Information Technology (BSIT)

Course: ITEP 414 – System Administration and Maintenance

Week: 1

Duration: 5 Hours (2 Hours Lecture + 3 Hours Laboratory)

Submission Type: Individual

Submission Platform: GitHub Classroom / GitHub Repository / LinkedIn

Laboratory Title

Building Your Professional System Administration Workspace

Overview

A System Administrator is expected to maintain organized documentation, collaborate using version control, continuously learn through professional platforms, and showcase technical accomplishments. In this laboratory, students will establish the complete professional environment that will be used throughout the semester.

By the end of the semester, every activity, documentation, script, configuration, and project completed in this course will become part of the student's professional portfolio that can be presented to employers, internship coordinators, and future clients.

Learning Objectives

At the end of this activity, the students should be able to:

Knowledge

1. Explain the importance of professional development platforms used by IT professionals.
2. Identify the software tools commonly used in System Administration.
3. Describe the role of Git and GitHub in version control and portfolio development.

Skills

Students will be able to:

1. Install all required software development and virtualization tools.
2. Configure their computer for future laboratory activities.
3. Create professional online accounts for learning and collaboration.
4. Create and organize a GitHub portfolio repository.
5. Publish their first technical documentation using Markdown.
6. Create a professional LinkedIn post describing their learning journey.

Attitude

Students will demonstrate:

- Professionalism
- Organization
- Technical documentation skills
- Continuous learning
- Digital professionalism



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Expected Learning Outcomes

After completing this activity, students should have:

- ✓ A fully prepared workstation
- ✓ Professional online accounts
- ✓ GitHub portfolio repository
- ✓ Organized project folders
- ✓ Professional GitHub profile
- ✓ LinkedIn profile
- ✓ Week 1 documentation

Required Software

Every student must successfully install the following software before proceeding with future laboratory exercises.

Software	Purpose
Git	Version Control
GitHub Desktop	Git GUI
Visual Studio Code	Code Editor
VirtualBox	Virtual Machine Platform
VMware Workstation Player (Optional)	Alternative Virtualization Platform
Ubuntu Desktop/Server ISO	Linux Operating System
Windows Evaluation ISO	Windows Virtual Machine
Google Chrome / Microsoft Edge	Browser

Required Accounts

Every student must create the following professional accounts.

Platform	Required
GitHub	✓
LinkedIn	✓

IN-CLASS ACTIVITY

Activity 1

Professional Software Installation

Objective

Prepare the computer for all laboratory activities throughout the semester.

Instructions

Install the following applications.

1. Install Git
2. Install GitHub Desktop
3. Install Visual Studio Code
4. Install VirtualBox
5. (Optional) Install VMware Workstation Player
6. Download Ubuntu Desktop or Ubuntu Server ISO
7. Download Windows 11 Enterprise Evaluation ISO

Verify Installation

Open each software and take screenshots showing that it launches successfully.

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The screenshots must clearly display:

- Software Name
- Version (if available)
- Student Name written in Notepad beside the software window

Required Screenshots

- Git
- GitHub Desktop
- VS Code
- VirtualBox
- Ubuntu ISO
- Windows ISO

Save all screenshots inside
Week01/screenshots/

Deliverable

Week01/

├── screenshots/
└── README.md

Activity 2

Professional Account Creation

Objective

Create professional online accounts that will be used throughout the course.

Instructions

Create the following accounts.

GitHub

Complete the following:

- ✓ Professional profile picture
- ✓ Full Name
- ✓ Short Bio
- ✓ Location
- ✓ Email (optional)
- ✓ Personal Website (optional)

LinkedIn

Complete

- ✓ Professional Photo
- ✓ Headline
- ✓ About Section
- ✓ Education
- ✓ Skills

Verification

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Take screenshots showing your profile pages.
Store screenshots inside
Week01/accounts/

Deliverable

Week01/
|— accounts/

Activity 3

Create Your Professional Portfolio Repository

Objective

Create the GitHub repository that will serve as your semester-long professional portfolio.

Repository Name

BSIT-SystemAdministration-Portfolio

Repository Description

Example

A collection of weekly laboratory activities, documentation, and projects completed in ITEP 414 – System Administration and Maintenance.

Repository Structure

Create the following folders.

BSIT-SystemAdministration-Portfolio

```
|
|— Week01
|— Week02
|— Week03
|— Week04
|— Week05
|— Week06
|— Week07
|— Week08
|— Week09
|— Week10
|— Week11
|— Week12
|— Week13
|— Week14
|— Week15
```

Commit and push the repository to GitHub.



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Bring-Home Assignment

GitHub Professional Profile

Update your GitHub profile with the following information:

- Professional profile photo
- Biography
- Skills
- Contact information (if you are comfortable sharing it)
- Custom profile README

Create a Profile README

Your GitHub Profile README must contain:

- ✓ About Me
- ✓ Career Goals
- ✓ Current Skills
- ✓ Technologies I Want to Learn
- ✓ Certifications
- ✓ Academic Projects
- ✓ Contact Information

Use Markdown formatting and include appropriate headings, bullet lists, and links.

Week01 Documentation

Inside the Week01 folder, create a file named:

README.md

Use the following template.

Week 1 – Building My Professional Environment

Student Information

- ✓ Name:
- ✓ Course:
- ✓ Section:
- ✓ Date:

Objectives

(Write your own learning objectives.)

Software Installed

List every software installed.

Professional Accounts

GitHub:

LinkedIn:

Installation Screenshots

Insert screenshots.

Challenges Encountered

Describe at least three problems encountered during installation or account setup and explain how you solved them.

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Reflection

Write a reflection (200–300 words) discussing what you learned during this activity and how these tools will support your development as a future System Administrator.

References

List the official websites and documentation you used.

LinkedIn Activity

Professional Learning Reflection

Create your first LinkedIn post summarizing your Week 1 experience.

Your post should include:

- A brief introduction about yourself.
- What you learned during the first week.
- The tools and platforms you installed.
- Your goals for the course.
- One screenshot of your completed workstation or GitHub repository.
- Relevant hashtags (e.g., #SystemAdministration #BSIT #GitHub #Linux #LearningInPublic).

Submission Checklist

Submit the GitHub repository link through your Learning Management System (LMS).

Your repository must contain:

- Week01 folder
- README.md
- Installation screenshots
- Account screenshots
- Organized folder structure
- GitHub Profile README (visible on your profile)

Also submit:

- LinkedIn post URL
- GitHub profile URL



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Grading Rubric (100 Points)

Criteria	Excellent	Good	Fair	Poor	Points
Software Installation	All required software installed correctly and verified with complete screenshots.	One minor issue or missing screenshot.	Multiple installation issues or incomplete verification.	Most required software missing or not functional.	25
Professional Accounts	All required accounts created, profiles completed professionally, and screenshots provided.	One profile incomplete or minor omissions.	Several incomplete profiles or missing screenshots.	Most accounts not created or incomplete.	20
GitHub Portfolio Repository	Repository created correctly with proper name, structure, commits, and organization.	Minor structural or naming issues.	Repository incomplete or poorly organized.	Repository missing or inaccessible.	20
Week01 Documentation (README.md)	Complete, well-written, properly formatted in Markdown, includes all required sections, screenshots, reflection, and references.	Mostly complete with minor formatting or content issues.	Missing multiple required sections or weak documentation.	Documentation incomplete or not submitted.	20
Professional Branding & LinkedIn Post	GitHub profile fully updated, LinkedIn profile completed, professional Week 1 post published with screenshot and hashtags.	Minor omissions in profile or post quality.	Incomplete profile or weak LinkedIn post.	Profiles not updated or post not submitted.	15
				TOTAL	100 Points

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Academic Integrity

All submissions must be the student's own work. Copying documentation, screenshots, repositories, reflections, or LinkedIn posts from classmates or online sources without proper attribution constitutes academic dishonesty and will be evaluated in accordance with the University's academic integrity policies.

Portfolio Milestone

At the end of **Week 1**, every student should have established a professional digital identity consisting of:

- A configured development workstation.
- A public GitHub repository for the course.
- A professional GitHub profile with a customized README.
- Professional learning accounts (GitHub, LinkedIn, Cisco Networking Academy, Microsoft Learn).
- A documented Week 1 laboratory activity in Markdown.
- Their first public LinkedIn learning reflection.

This milestone lays the foundation for the remaining weeks of the course, where each laboratory activity and project will be added to the portfolio, resulting in a comprehensive showcase of practical System Administration skills by the end of the semester.